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Differentiation between AI and human responses to research ethics and integrity cases– implications for teaching

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Abstract

The current study investigates the differences between AI-generated and human responses to ethical analyses in research ethics and integrity education, using the SOLO taxonomy to evaluate the depth of understanding. As generative AI models increasingly produce human-like responses, concerns arise regarding their impact on academic integrity and critical thinking. The findings reveal that while AI can achieve multistructural and relational levels of understanding, human responses, particularly from experts, consistently demonstrate higher levels of critical thinking and integration of ethical principles. AI responses often lack the depth and novelty characteristic of human analyses, particularly in proposing innovative solutions and considering broader implications. This suggests that reliance on AI in ethics education may hinder the development of essential skills needed for academic and professional success. The study also highlights the potential for AI to serve as a tool for evaluating student responses to learning assignments, provided transparency is maintained. Overall, educators are encouraged to design assessments that foster independent thought and creativity while integrating AI responsibly into the learning process. Future research should explore the long-term effects of AI on ethics competencies and the effectiveness of various AI platforms in educational settings.

Keywords Differentiating between human and AI generated responses, Ethics and integrity education, AI impact on ethics education

Introduction

Generative AI models have demonstrated strong performance in producing coherent and human-like responses for academic assessments. This raises important questions about the authenticity of student work, and teachers need to make an extra effort to see what the learners are learning and if they achieve their intended learning outcomes. Concerns about the use of Generative AI and its implications have been raised in higher education (e.g. Kirwan 2024; Mennalla and Quadros-Mennalla 2024; Park and Milner 2025; Veras et al. 2024). Nevertheless, we completely agree with Eaton (2025) who indicates that we should not consider AI as the ‘bad guy’ that should be prohibited



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from educational contexts— AI is here to stay and we as educators should become familiar with the possibilities and drawbacks of its use to plan learning and teaching more effectively.

Research ethics and integrity education utilises various methods like lectures, case studies, discussions, and practical exercises to help students understand the importance of originality and the ethical use of information in their work (i.e. Löfström and Tammeleht 2023). In the era of AI various challenges may arise, for example, the potential for students to rely too heavily on AI-generated content, leading to a decline in critical thinking and creativity. Indeed, teachers already have significant concerns over the reliance on AI tools which may diminish students' abilities to form original arguments or critically evaluate information (O'Sullivan 2025).

Such reliance could ultimately hinder the development of essential skills necessary for academic and professional success, prompting educators to seek innovative strategies that integrate AI responsibly while fostering independent thought and creativity. Moreover, teachers need to reconsider tools used for evaluating learning in ethics education (to maintain the credibility of knowledge attainment) and beyond (to elicit ethical behaviour in the research community). These tools must adapt to ensure they assess not only knowledge transmission but also the learners' ability to engage with complex ethical dilemmas in a nuanced manner, encouraging students to think critically and independently. For example, assessment techniques, such as oral examinations and reflection diaries, could be used to better gauge a student's understanding and critical thinking skills beyond what generative AI can produce. Some training formats may consider re-introducing on-site proctored examinations.

Prior studies (Williams 2024; Chaudhry et al. 2023; Currie et al. 2023; Govender 2024; Maeda 2024) consistently indicate that AI tools, despite their potential, face significant challenges in tasks requiring abstract reasoning, critical analysis, and complex problem-solving, which are all essential aspects in developing ethics competencies. This indicates the need to rethink educational strategies - both teaching methods as well as evaluation - that prioritize human judgment and creativity, ensuring that students are equipped with the skills necessary to navigate an increasingly automated world while maintaining their ethical compass. A comprehensive study by Weber-Wulff et al. (2023) tested fourteen detection tools for AI-generated texts. They concluded that the detection tools are neither accurate nor reliable diagnosing human-written documents as AI-generated, and vice versa. These findings put additional pressure on educators to develop methods for the evaluation of learning. Proctored exams are one means, however, there is room for development of reflection-based ways of evaluating what has been learned, especially at the higher levels of understanding (see Richards et al. 2024).

We are not aware of empirical studies conducted in the context of research ethics and integrity (REI) education pertaining the use of AI and how its use can be detected. In our study we were interested in how AI generated responses to ethics analysis questions compare to answers provided by human learners. Our aim was to assess not only the quality of these responses but also to understand the implications for teaching and learning of research ethics and integrity.

Literature overview on research on the use of AI tools in assessed assignments

We reviewed findings from recent studies on the use of AI tools in completing the assignments and the implications for achieving learning outcomes. Various studies have been conducted to see how well AI performs in completing tasks in HE context. AI studies appear to convey similar results about the failure of AI to reach responses on higher cognitive levels. Yeadon and Hardy (2024) explored the use of OpenAI's GPT-3.5-turbo model in physics education, assessing its performance across different academic levels. The model scored an average of 83.4% on GCSE (*General Certificate of Secondary Education*— lower-secondary school-leaving exams in the UK), 63.8% on A-Level (upper-secondary school leaving exams in the UK), and 37.4% on university-level physics exam questions, revealing a decrease in performance with increasing question complexity. The study highlighted AI's limited mathematical capabilities, with only 45.2% accuracy in mathematical operations, suggesting a need for educators to carefully evaluate AI's role in coding and essay assignments (Yeadon and Hardy 2024). Exam items was the focus also in the research by Maeda (2024), who introduced a method for field-testing exam items using AI, which showed a high correlation (0.82) with true response probabilities, though AI responses did not match human accuracy. The analysis revealed that while AI tools like Microsoft Bing, ChatGPT, and Google Bard can provide accurate answers for simpler tasks, they face challenges in handling questions that demand higher cognitive engagement, leading to incomplete responses in 76.2% of cases (Maeda 2024). Moreover, Govender (2024) assessed the effectiveness of the mentioned AI tools on Bloom's taxonomy of learning. While AI tools performed well on lower cognitive level questions, they struggled with higher mark-weighted questions, indicating limitations in handling complex cognitive demands (Govender 2024).

Williams (2024) evaluated the performance of generative AI tools on essay questions in biomedical sciences. The results indicated that AI-generated responses often lacked scientific accuracy and detailed mechanistic understanding, which are critical in fields like biomedical sciences indicating a need for caution when relying on AI for complex academic tasks (Williams 2024). The results also showed that while AI performed well on lower-level questions, it struggled with higher-level questions that demanded critical thinking and in-depth analysis, highlighting the varying levels of response quality (Williams 2024). Similar results have been identified by Richards et al. (2024), who propose the use of reflection-based assignments to encourage student learning.

Prior findings point to important questions about the role of AI in education and whether AI can truly supplement human expertise, especially in disciplines that require nuanced understanding and analytical skills. The implications of these findings suggest that while AI can serve as a valuable tool for basic information retrieval and support, it may not yet be capable of fully replacing the depth of insight and critical evaluation provided by trained professionals in complex academic fields.

Tammeleht and Löfström (2025) looked into measurement tools used in REI training in the past two decades and assessed if and how AI could potentially be used by learners to provide responses (we do not consider only cheating). We could identify various tests used to evaluate learning outcomes in REI training, like DIT (*Defining Issues Test*) (Rest et al. 1999), TESS (*Test for Ethical Sensitivity in Science*) (Clarkeburn 2002), DEST (Bebeau et al. 1985), EDM (*Ethical Decision-Making*) (Mumford et al. 2006), etc. There were also other measures to collect learning outputs that could give insights to teachers

on how the learners were advancing, for instance, vignettes, learning logs, group portfolios, etc. (Authors 2025). Research indicates that AI can simulate moral reasoning by analysing dilemmas and generating responses. For example, researchers tested LLMs like GPT-4 using DIT-style prompts and found coherent moral reasoning in models like ChatGPT and GPT-4, also smaller models like *LlamaChat* performed well (Tanmay et al. 2023). In the study (Tanmay et al. 2023), AI tools were evaluated using the *Defining Issues Test* (DIT) to measure their moral reasoning abilities. The results showed varying performance across different models. Overall, GPT-4 showed the highest moral development score, while other models performed at levels comparable to adult humans or college students. However, inconsistencies were observed across different dilemmas, indicating gaps in understanding and reasoning abilities. (Tanmay et al. 2023). There are examples of AI trained in recognizing ethical constitutions, such as the *United National Declaration of Human Rights*, and constitutional language (see Sharples 2023), but the values it is taught to recognise, such as liberty and personal security, may not necessarily coincide with ethical principles in all situations.

Thus, various ethics tests can be completed with the use of AI, but the results may be controversial. Overall, we can assume that AI can support students by providing practice scenarios, structuring responses, or analysing data, but it cannot ethically replace human input in assessments designed to measure personal moral reasoning, sensitivity or growth.

We are not aware of studies conducted on the quality of AI responses and their comparison to human responses in the context of research ethics and integrity education. To investigate the quality of AI responses and compare them to human responses we asked various AI platforms to conduct ethical analysis on ethics cases, evaluated the responses based on the SOLO (*Structure of the Observer Learning Outcomes*, see explanation in Method section) taxonomy (Biggs 1996; Biggs and Tang 2007) and compared the levels of understanding to the human responses. We proposed the following research questions:

- How do AI-generated ethical analyses measure up against those produced by human learners based on the SOLO taxonomy?
- What insights can be drawn from this comparison regarding the potential use of AI to methods used in ethics education?

The findings aim to give a snapshot of the strengths and weaknesses of AI in ethical reasoning, ultimately contributing to a better understanding of how the use of these technologies may impact ethics education.

Method

The current study used qualitative research methods to analyse the responses generated by AI platforms alongside those of human learners. Table 1 provides an overview of the research method:

We asked 2 AI platforms to analyse ethics cases and then analysed the responses based on the SOLO taxonomy matrix designed for ethics context (Tammeleht et al. 2019). We compared the results to our previously collected responses by human learners. Human learners participated in a training that utilised case-based collaborative learning format— this means learners worked in groups and discussed ethics cases following

Table 1 Overview of the research method

Method	Qualitative deductive content analysis
Participants/tools	AI platforms: <i>Huggingface</i> , <i>Perplexity</i> Human learners: 3 groups in case-based collaborative learning format Ethics expert: 1 individual response in Word document
Data collection and analysis	AI platforms: were asked to analyse ethics cases using standardized questions (ethical analysis) Human learners: group analysed ethics cases (ethical analysis), discussions were recorded, transcribed. The expert worked individually, provided the analysis of the ethics case in writing. Analysis method: Responses were evaluated using the SOLO taxonomy by Brisk application and a human coder.

ethical analysis steps (the same format of training is still in use in various institutions). The group discussions were recorded and transcribed, the unit of analysis for learning outcomes was a group (Authors 2022). The only individual response was provided by the expert who was asked to provide the response in a Word document. A human facilitator evaluated the written responses and group discussion transcripts. Then the same responses were transferred to an AI platform, the evaluation matrix was inserted, and the evaluation was collected. This could be interpreted as using a second coder of data.

We tested two platforms (*Huggingface* and *Perplexity*) to prompt the responses similarly to how the human learners have been prompted. We provide a case with an ethical dilemma (see appendix for cases used as part of this study) and provided questions for ethical analysis (from Mustajoki and Mustajoki 2017) (both human respondents and AI platforms got the same set of questions exactly the same way): *Which ethical principles may be present in this case? Who are the stakeholders? Why them? What are the responsibilities and rights of the stakeholders? What are the possible courses of action? What are their implications?*

Huggingface and *Perplexity* were chosen as they are free applications, and they have not been tested as much as e.g. ChatGTP or Microsoft Bing (concluded based on previous literature). In addition, we had tested/ piloted ChatGTP before and did not receive responses beyond multistructural level, so we were hoping to receive better quality answers. Also, *Perplexity* is one application recommended (<https://www.perplexity.ai/hu/b/blog/getting-started-with-perplexity>) to HE students and researchers, so it is probable , that learners in HE context might use it.

This approach aimed to identify patterns and differences in levels of understanding displayed in AI and human responses. We also analysed the reasoning, depth of understanding, and the overall quality of ethical analyses produced by AI compared to human learners. We also tested *Brisk* application to provide evaluation to both AI generated as well as human responses and compare how they match the evaluation provided by a human teacher.

Data were collected as text in the Word document, and we used deductive content analysis to identify the levels of understanding. We also looked at the process the AI and human learners used to analyse the case. Data were analysed based on the SOLO taxonomy rubrics created for ethics context, evaluators were an ethics training facilitator and an AI application *Brisk* (AI teaching assistant for teachers, which in addition to lesson planning and content creation can support teachers to evaluate learning outputs and help provide feedback <https://www.briskteaching.com>).

The SOLO (*Structure of Observed Learning Outcomes*) taxonomy is an educational framework that categorizes learning outcomes based on the complexity and depth of understanding achieved by learners (Biggs 1999). In the context of research ethics and integrity education, the SOLO taxonomy serves as a valuable tool for assessing and enhancing students' comprehension and ethical sensitivity (Tammeleht et al. 2019).

The SOLO taxonomy rubrics can be outlined as following for the REI context (based on Tammeleht et al. 2019), and they were the basis of evaluation in this study:

Extended abstract level of understanding

The learner goes beyond conceptualizing the present issue making steps towards relating the ethical issues to applications beyond the present case. Displaying the ability to theorise, generate, generalise, hypothesise, create or reflect.

Relational level of understanding

The learner displays an ability to address the most relevant ethical issues and provide explanations pointing out interrelations and providing examples demonstrating own reasoning.

Multistructural level of understanding

The learner demonstrates that concepts had been understood appropriately (by naming them), but struggles to make connections between them and to draw conclusions based on interrelations. Knowledge-telling approach, no structuring.

Unistructural level of understanding

The learner identifies one relevant aspect displaying some familiarity with relevant concepts, but failing to address some more pertinent dimensions of the case.

Prestructural level of understanding

The learner fails to identify a relevant ethical perspective or does not approach it in a meaningful way, repeating the words in the case without displaying evidence of own processing.

Research ethics

The current research followed the Estonian national code of conduct (Hea Teadustava 2023), the Finnish National Board on Research Integrity guidelines (2023) as well as the all-European code of conduct for research integrity (ALLEA 2024). Data by human participants was collected during another study (published by Authors 2022) and were generated as part of REI training, participation or non-participation had no impact on passing the training. An ethics review was not required for that research, in which participants were adults who gave their consent voluntarily, and which addressed neither sensitive topics nor interfered with the physical integrity of participants. Consent for using the written response and group discussion content for research purposes was sought prior to data collection (an information letter and a signed consent form). Consent was not sought from AI (as AI does not understand what it consents to).

We outline how AI tools were utilised in this study. All applications used are openly accessible (excepting *Curre*) and free of charge (even though this puts word- or inquiry-limits to the users). We used the following applications:

1. *Consensus* (and *Zenodo*)– we used the app to find additional articles on the topic (we had identified and read some articles on our own). The suggestions provided by *Consensus* were transferred to *Zenodo* and articles were read by us, the final selection of articles we decided to use for the literature overview was made in *Zenodo*.
2. *Perplexity*– we used the application for specifying research focus and the bigger context of the study. Occasionally, some specifying questions to boost writing were also inserted into *Perplexity*.
3. *Curre* (the University of Helsinki internal AI system) was used for generating tables and the initial overviews of tables (that were then modified by the authors). *Curre* was also used to help shorten the article summaries and comparisons, and to generate the initial version of the abstract.
4. *Brisk* application was used as a second ‘coder’– based on the SOLO rubrics, ethical analyses by humans and AI platforms were evaluated and feedback provided by the app was also collected.
5. *SciSpace* was used to support writing the article. The app provided a generic outline for the article and provided ideas for starting and continuing thoughts (even though most of them were ignored as the text was too generic or not pertaining to the research topic). It should be noted that *Research Ethics* section was not included in the generic structure provided by AI.

Results

We used 4 different cases (see topics in Table 2 and descriptions in the appendix) in total and ethical analysis steps to collect responses from 2 AI platforms, 1 ethics expert and 3 human groups, the results are displayed in Table 2.

As indicated in Table 2, the evaluations of both the facilitator and the AI application coincided to a large extent. In case there are two levels indicated by evaluators,

Table 2 Summary of levels of understanding (SOLO levels) as evaluated by the facilitator and the AI evaluator

Cases	Respondents	Facilitator evaluation	AI tool evaluation
Case 1– data collection with children, but some parents do not give consent	human (expert)– written response	Relational/ extended abstract	Extended abstract
Case 2– supervisory neglect to a PhD student	human group 1– group discussion	Multistructural/relational	Multistructural/ relational
Case 3– potential misappropriation of PhD student ideas by the supervisor	human group 2– group discussion	Relational	Multistructural/ relational
Case 4– potential deviance of ethical research procedures	human group 3– group discussion	Relational/ extended abstract	Relational/ extended abstract
Case 1– data collection with children, but some parents do not give consent	AI 1 (Huggingface)– written response	Multistructural	Multistructural
Case 1– data collection with children, but some parents do not give consent	AI 2.1 (Perplexity 1)– written answer	Multistructural/ relational	Multistructural
Case 1– data collection with children, but some parents do not give consent	AI 2.2 (Perplexity 2)– written answer	Relational	Relational/ extended abstract

the responses mostly displayed the first level but in case of one or two questions the other level was displayed (so the more prevailing level would be the first one indicated in Table 2). In Case 1, an expert human provided a written response, which was evaluated as relational/extended abstract by the facilitator and extended abstract by AI evaluator. In Cases 2, 3, and 4, groups of humans engaged in discussions, with varying levels of understanding observed. Case 2 was evaluated as multistructural/relational by both human and AI evaluators. Case 3 saw a facilitator rating of relational, while AI rated it as multistructural/relational. Case 4 received a consistent evaluation of relational/extended abstract from both evaluators. In Case 5, an AI 1 provided a written response, rated as multistructural by both evaluators. Case 6 was completed by AI 2.1 and was rated as multistructural/relational by the facilitator and multistructural by AI evaluator, and Case 7 completed by AI 2.2 was rated as relational by the facilitator and relational/extended abstract by AI evaluator.

In the responses exhibiting a relational level, the content of the AI-generated responses was correct, e.g., naming beneficence, non-maleficence and respect for persons (e.g. Table 3 Sect. 1). However, AI failed to connect autonomy and respect for persons, which could easily be argued for as being related. Likewise, the stakeholders listed (Table 3, Sect. 2), e.g., children, parents and teachers, are correct answers, but on a relational level one would expect some elaboration of the rights of the child (as in human-generated response), and how there could be a discrepancy between a guardian's indicated consent or lack of it and their child's wish in the moment of data collection. Many other relationships among stakeholders could be identified.

To illustrate the allotted levels and content of the responses (for the same case 1, see the topic in Table 2 and the description in the appendix) we provide an example of a response analysis for AI 1 response to a case in Table 3 and for human respondent (expert) in Table 4.

The levels of understanding at various stages of ethical analysis and comments provided by the facilitator and the AI application were analysed to gain insights into the ethical analysis process and make comparisons of the various responses.

First, the levels of understanding of AI responses were monitored. AI 2.1 demonstrated understanding of a multistructural level, with suggestions from the AI evaluator to make connections between different aspects more explicit to reach a relational level. AI 1 showed understanding primarily on the multistructural level, with advice from AI evaluator to connect elements more deeply to achieve a relational understanding. AI 2.2 displayed a relational level of understanding, with suggestions to theorize about underlying principles to potentially touch upon an extended abstract level.

Human responses indicated the following. Human expert displayed an extended abstract level across all areas. Human group 1 showed understanding between multistructural and relational levels, with some elements reaching extended abstract (courses of action and their implications). Human group 2 was primarily at multistructural and relational levels, while human group 3 displayed understanding at multistructural and relational levels, with aspects reaching extended abstract (again the courses of action and their implications).

Similarities are that both AI and human responses demonstrate knowledge of ethical principles and stakeholders. Feedback for both groups often include recommendations to deepen analysis by exploring interactions, conflicts, and broader implications. There

Table 3 Example response 1 by AI 1 and evaluations by a human facilitator and AI evaluator

Topic	Content	Facilitator evaluation	AI tool evaluation
1. Ethical principles	Respect for persons: The children who were removed from the room may feel punished, excluded, or stigmatized, which could harm their self-esteem and dignity. Beneficence: The researchers have a responsibility to ensure the well-being and safety of all children involved, including those whose parents did not provide consent. Non-maleficence: The removal of some children from the room may cause them emotional distress, which could be considered a harm. Autonomy: The children who were removed from the room may not have had a choice in the matter, and their parents' decision not to provide consent may not have been fully explained to them.	Multistructural– knowledge-telling approach, no connections, no examples.	Multistructural: • You identified relevant ethical principles, including respect for persons, beneficence, non-maleficence, and autonomy. • Consider exploring how these principles interact and potentially conflict in this scenario to deepen your analysis. • The principle of justice could also be relevant here - consider how it relates to the other principles you discussed.
2. Stakeholders, their responsibilities and rights	Children: The preschool children who are participating in the research, including those whose parents provided consent and those who did not Parents: The parents who provided consent for their children to participate in the research, as well as those who did not. Researchers: The research team, including the principal investigator and any assistants. Preschool teachers: The teachers who are responsible for the children's care and education during the research activity.	Multistructural– no connections. Also, not seeing a bigger picture, a much wider circle will be influenced.	Multistructural: • Your stakeholder analysis includes key groups: children, parents, researchers, and preschool teachers. • To strengthen this section, examine how these stakeholders' interests and roles interconnect. • Consider expanding your analysis to include potential indirect stakeholders, such as the preschool administration or ethics review board.
	Children: The right to be treated with respect, dignity, and care. The researchers and preschool teachers have a responsibility to ensure their well-being and safety. Parents: The right to make informed decisions about their child's participation in research. The researchers have a responsibility to provide clear information about the research and obtain informed consent. Researchers: The responsibility to conduct research in an ethical manner, ensuring the well-being and safety of all participants. They also have a responsibility to respect the autonomy of parents and children. Preschool teachers: The responsibility to care for and educate the children, as well as to support the researchers in conducting the research in an ethical manner.	Multistructural– rights recognised for the identified stakeholders, no connections, no examples.	Multistructural: • You outlined key rights and responsibilities for each stakeholder group. • To reach a more relational level of understanding, analyze how these rights and responsibilities might conflict or complement each other. • Consider how the responsibilities of one group might impact the rights of another.

Table 3 (continued)

Topic	Content	Facilitator evaluation	AI tool evaluation
3. Possible courses of action, their implications	Modify the research design: Consider alternative methods that do not require separating children based on consent, such as using a different data collection method or involving all children in a way that does not require consent. Provide alternative activities: Offer alternative activities for the children who were removed from the room, to minimize feelings of exclusion and ensure their well-being. Improve communication: Ensure that parents and children understand the research and the reasons for separating the children. Provide clear information about the research and the benefits and risks of participation. Reconsider the consent process: Reflect on the consent process and consider whether it was adequate. Consider obtaining consent from parents in a way that is more inclusive and respectful of their autonomy. Modified research design: May require significant changes to the research plan, potentially affecting the validity and reliability of the data. Alternative activities: May require additional resources and planning, but could help minimize harm to the children who were removed from the room. Improved communication: May help build trust with parents and children, but may not fully address the ethical concerns. Reconsidering the consent process: May require a re-evaluation of the research plan and the consent process, potentially leading to changes in the way consent is obtained in the future.	Multistructural - While all the provided courses of action sound plausible, there is little connection to the overall topic. Very rule-based, little empathy. Multistructural– not seeing the wider picture, knowledge telling approach.	Multistructural: Your proposed courses of action show consideration of ethical alternatives. • To enhance this section, discuss how these actions might be combined or prioritized. • Analyze how each action might address (or fail to address) the concerns of multiple stakeholders simultaneously. Multistructural: • You provided an initial analysis of implications for each course of action. • To deepen your analysis, consider how these implications might interact or create cascading effects. • Explore potential unintended consequences and how they might affect different stakeholders in interconnected ways.
4. Overall		Multistructural– decent recognition of various aspects, no connections or seeing the bigger picture– especially the wider impact of the situation. No examples, no empathy.	Multistructural: • Your response demonstrates understanding of individual elements, but more connections between these elements are needed to reach a fully relational level. • To improve, try to draw more explicit links between different aspects of your analysis, such as how ethical principles inform stakeholder responsibilities, or how courses of action relate to specific ethical concerns. • Consider formulating some overarching principles or insights that emerge from the interconnections in your analysis.

Table 4 Example response 2 by a human respondent (the expert) and evaluations by a human facilitator and AI evaluator

Topic	Content	Facilitator evaluation	AI tool evaluation
1. Ethical principles	Ethical issues that could emerge in this case: underage children, parental consent, procedure of getting the consent, considering the wishes of the child, organising data collection, cooperation with the (pre)school, which is more harmful for the child - being recorded (for the research purposes) or being upset about the procedure of fulfilling the parent's demands. Ethical principles in this case: respect for autonomy - in this case there is a conflict between the autonomy of the child and the autonomy of the parent - for the researchers there is no simple answer. The procedure of informing should be altered to prevent these situations from happening in the future. doing no harm (non-maleficence) - in this case the researchers' only chance of not harming the children whose parents had not given their consent would have been not proceeding with data collection, but in that case they would be harming the research/society/greater good. benefiting others (beneficence) - the research is probably important to the society, but it is not right to harm anyone to benefit others. being just (justice) - the researchers must be fair towards the children, their parents, the school, the society - if their rights get into a conflict of interest, new means of data collection must be found. being faithful (fidelity) - researchers must respect the wishes of the parents, but also children.	Relational - good recognition of potential ethical aspects, ethical principles connected and exemplified.	• You demonstrate an Extended Abstract understanding by identifying multiple relevant ethical principles and applying them to this complex situation. • Your explanations show how these principles interact and potentially conflict in this scenario, indicating high-level synthesis. • You've gone beyond simple identification to explore the nuances and implications of each principle in this context.

Table 4 (continued)

Topic	Content	Facilitator evaluation	AI tool evaluation
2. Stakeholders, their rights and responsibilities	Researchers - the team of researchers need data for their research, they have planned their activities and data management, they have the support from their leader and institution, they need the research to bring new knowledge into the society and also promote their own careers. Children - research may often include children, they will also benefit from the research. By law, underage children (there are some differences of age in different countries) need a parental consent to be part of the research. At the same time, UN Article 12 states that the child's opinion must be asked and considered (depending on their development). If the parent's and child's opinions contradict, the researchers cannot decide what to do, the best option might be to quit data collection and find new measures, and plan the informing procedure better. Parents - are responsible for their children, have the right to decide for their underage children, must consider the well-being of their children. Parents should be aware of the implication of their decision - whether their decision may harm the child (mentally), whether they are hindering the improvements in the society. Parent should seek for more information to consider all the alternatives. (Pre)School - if the school allows the researchers conduct data collection in their institution, then the school leader should also be informed and evaluate the situation - either make suggestions on how better organise data collection and what the benefit is to the children/school. Research institution - provide guidance, training to the researchers on how to better organise data collection, especially involving people/children. Maybe an ethics review is necessary or advice from the ethics committee. Society - needs research for improvements, should encourage researchers conduct research to develop better policies. It is important to spread knowledge that all citizens can contribute to improvements by participating in surveys.	Relational - seeing stakeholders, seeing their connections and a bigger picture. Referring to specific documents.	• Your stakeholder analysis reflects Extended Abstract thinking by considering a comprehensive range of stakeholders and their interconnected roles. • You've thoughtfully examined the responsibilities and rights of each group, showing an ability to see multiple perspectives. • Your analysis goes beyond listing stakeholders to exploring potential conflicts and balancing of interests.

Table 4 (continued)

Topic	Content	Facilitator evaluation	AI tool evaluation
3. Possible courses of action, their implications	This event is done and it was harmful to the children - it would have been better if the researchers had stopped the data collection procedure once they learned that some children cannot participate in video recording. Maybe more time should have been spent on informing parents. In the future the research teams should plan more time (e.g. in the data management plan) on informing and getting the consent - organising an information seminar for the parents, encouraging questions and asking also the children, giving families more time to make the decision, inventing new ways to organise recordings. Parents should also give it more thought - how can they contribute to research and society, maybe they could read some newspaper articles about how research is conducted and what happens to the collected data. Research institution should organise trainings and guidelines on data management, collection and protection.	Extended abstract - empathy, seeing the current status and the future possibilities. Examples of how to prevent this form happening in the future, examples. Also how various stakeholders may influence the situation.	• Your discussion of possible courses of action demonstrates Extended Abstract understanding by considering both immediate and long-term implications. • You've shown the ability to generate novel solutions and critically evaluate potential outcomes. • Your suggestions for future improvements indicate an ability to transfer learning from this scenario to broader research practices.
4. Overall		Relational / extended abstract	• Your response consistently demonstrates Extended Abstract thinking across all areas of analysis. • You've shown the ability to integrate multiple concepts, consider abstract principles, and generate novel insights and solutions. • Your analysis goes beyond the immediate case to consider broader implications for research ethics and societal impact.

is a consistent call across both types of responses to make more explicit connections between ethical principles, stakeholder interests, and proposed actions.

The AI and human responses also indicated differences. There seemed to be consistency in higher levels of understanding - human responses, particularly from the expert, consistently reached higher levels of understanding (relational and extended abstract). While AI responses were varied only one AI respondent (AI 2.2) consistently reached relational levels.

The potential for extended abstract thinking can also be considered a difference. Human groups, especially the expert, were noted for achieving an extended abstract level more frequently. AI responses, except for AI2.2, generally show potential but remain at multistructural or relational levels. This may be connected to the depth and novelty in responses: human responses can be highlighted for their potential to suggest novel solutions and consider long-term implications, aspects less emphasised in AI responses.

In summary, while AI responses show strong analytical capabilities, human respondents, particularly the expert, display a more consistent ability to operate at higher levels of understanding and to integrate broader and more novel perspectives into their analyses.

Discussion

In our study we aimed to find out how AI-generated assignment responses (i.e., ethical analyses of an ethics case) measure up against those produced by human learners based on the SOLO taxonomy. We hoped to gain insights from this comparison on the potential use of AI to teaching and evaluation methods used in ethics education.

Based on the data collected and using the SOLO taxonomy as a framework for comparison, we can deduce that AI-generated ethical analyses differ compared to those produced by human learners. AI 2.1 and AI 1 responses generally fell within the multistructural level of the SOLO taxonomy. The responses demonstrate the ability of AI to identify and describe various ethical principles and stakeholders but lack deeper integration and connection among these elements, which is necessary to reach higher levels of understanding. AI 2.2 responses were on a relational level making connections between ethical principles, stakeholder needs, and potential actions. The responses went beyond just listing or describing elements to explaining and analysing relationships and interactions. Still, as indicated by Tanmay et al. (2023), AI tools can use ethical analysis on a level suitable for adults. Indeed, the cases and ethical analysis tool used for collecting data in this study is intended for advanced level learners (learners who already have some knowledge on research ethics and integrity) (see Tammeleht and Löfström 2025). Prior studies also indicate that beginners of ethics studies generally remain solidly on the multistructural level (Tammeleht et al. 2019).

Human-generated responses usually displayed high levels of understanding. Human expert consistently operated at an extended abstract level, indicating a deep, conceptual understanding and the ability to generalize principles beyond the given scenario. This level involves synthesising information into coherent wholes and theorising about underlying principles or broader implications. Human groups 1, 2, and 3 generally showed understanding at the multistructural to relational levels, with instances approaching the extended abstract level (usually at the end of the task). Humans were capable of identifying and describing ethical issues and stakeholders and often made connections among

these elements. They occasionally demonstrated the ability to think abstractly and propose innovative or long-term solutions.

As the quality of AI vs. human responses in research ethics and integrity context has to our knowledge not been systematically evaluated previously, we hope to bring new insights to the wider discussion on the impact of AI to ethics education. We could see that human-generated analyses, particularly from the expert, tend to reach deeper levels of understanding more consistently. They are more likely to exhibit responses on an extended abstract level and to apply principles in broader contexts compared to AI-generated analyses. In contrast, AI responses, except for AI 2.2, generally hover at the multi-structural level, with less frequent exhibition of relational content. We may conclude that lower SOLO levels may serve as indications of AI being used by learners in training context. This may not always be the case, but teachers and trainers need to consider this option along with assignments and evaluation of learning, which encourage reflection.

Human learners show a greater propensity for suggesting novel and systemic solutions, reflecting a broader and more integrated understanding of ethical issues. AI-generated responses, while competent in identifying and analysing components, less frequently suggest innovative or comprehensive solutions that consider broader systemic implications. These findings are very consistent with prior research done in different fields (Chaudhry et al. 2023; Currie et al. 2023; Govender 2024; Maeda 2024; Richards et al. 2024; Williams 2024) indicating that while threshold level understanding is reachable to AI tools, the higher levels of understanding and applying the knowledge in novel context may not be in AI's reach yet. We anticipate that this is bound to change as technology develops. However, at this time, this knowledge could guide ethics teachers to create tasks that demand for considering the bigger picture, indicating connections, providing novel solutions and applying the gained information in different contexts. Using just knowledge tests or even well-tested ethics measuring tools (like DIT, TESS, EDM) may be prone to learners using AI to complete them and it may not be possible to distinguish who the respondent has been.

Another aspect we would like to point out is the usability of AI application for evaluating learner responses to case analysis. With a solid taxonomy (like SOLO) AI evaluator can evaluate group reports, oral presentations or written responses quite accurately and also add relevant comments on the content and how to improve the level of understanding. Nevertheless, for the sake of transparency, if the facilitator uses an AI evaluator, the learners should be notified of it.

Limitations and future research suggestions

Indeed, we are aware how fast AI tools are advancing and soon it may not be possible to distinguish between a human expert response from the AI response. In addition, the sample we used was small and future studies could make use of broader samples of various kinds in addition to the case-based learning assignments used here. Future studies could also use different AI platforms to collect responses from AI and evaluate the quality of those responses. There are numerous generative AI tools to be exploited, and we realise we have applied very few in this research. Furthermore, involving more experts may have strengthened the reliability of the interpretations. Future studies should also monitor the long-term effect on ethics competencies when AI is integrated in ethics education.

Conclusion

Prior studies have highlighted that while AI can assist with basic tasks, its limitations in depth and insight, coupled with risks for academic dishonesty, necessitate a strategic adaptation to AI technologies in education, and this requires understanding the context where AI tools could be used. AI in education offers promising benefits but also poses significant challenges, particularly in maintaining academic integrity and handling complex cognitive tasks. Educators must consider these factors to leverage AI's potential responsibly and effectively in educational contexts.

Our study indicated that while AI-generated ethical analyses are capable and sometimes reach relational levels of understanding, human-generated analyses, particularly from the expert, tend to demonstrate greater depth, complexity, and consistency in higher-level thinking according to the SOLO taxonomy. This includes a greater ability to integrate knowledge, think abstractly, and apply insights in broader or novel contexts. With these insights, facilitators of ethics training can create tasks and assignments that would require learners to display those competencies.

We hope that the insights from our study provide ethics teachers a taxonomy and strategy to create ethics training formats that are not so prone to unsolicited AI tampering that diffuses what learners are actually (not) learning. In addition, recognising benefits of the AI evaluation application in ethics context, we think that it provides teachers with a tool to lighten the workload of text-based tasks (like learning diaries, written reports, etc.). Of course, the same requirement to not tamper with AI naturally concerns teachers as well, and so transparency is key to nurturing trust in the pedagogical relationship.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s40979-025-00192-9>.

Supplementary Material 1

Acknowledgements

We would like to thank everyone who has participated in this study.

Author contributions

Conceptualization: EL; Data curation: AT; Formal analysis: AT, EL; Funding acquisition: EL; Investigation: AT, EL; Methodology: EL, AT; Project administration: EL; Resources: AT Software: AT; Supervision: EL; Validation: EL; Visualization: AT; Writing—original draft: AT; Writing—review & editing: EL, AT.

Funding

Open Access funding provided by University of Helsinki (including Helsinki University Central Hospital). Open Access funding provided by University of Helsinki (including Helsinki University Central Hospital). This research was funded by BEYOND BAD APPLES project, which has received funding from the EU's Horizon Europe program under grant agreement No 101094714.

Data availability

The current paper uses data generated by AI applications. The content generated by humans comes from a small sample and making it openly accessible may pose a threat to anonymity of participants. Aggregated data can be made accessible upon request (please contact the corresponding author).

Declarations

Ethics approval and consent to participate

Not applicable. Data by humans was collected as part of another study (published by Tammeleht and Löfström 2025). Ethics approval was not necessary for that research. An ethics review is required when research involves intervention in the physical integrity of participants; deviates from informed consent; involves participants under 15 being studied without parental consent; exposes participants to exceptionally strong stimuli; risks long-term mental harm beyond that encountered in daily life; or involves security risks (Finnish National Board on Research Integrity, 2019). None of these conditions materialised in the study, and consequently an ethics review is not expected in the Finnish context. Finnish National Board on Research Integrity TENK. (2019). The Ethical Principles of Research with Human Participants and Ethical

Review in the Human Sciences in Finland. https://tenk.fi/sites/default/files/2021-01/Ethical_review_in_human_sciences_2020.pdf.

Consent for publication

On behalf of both authors, the corresponding author states that there is no conflict of interest (no third parties being influenced by the article content). Both authors have approved the manuscript.

Competing interests

The authors declare no competing interests.

Received: 25 March 2025 / Accepted: 11 June 2025

Published online: 14 July 2025

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